Transformers for Image Recognition at Scale: Vision Transformers and Data Regime Dependence

1. Introduction

Deep learning for computer vision has long been dominated by convolutional neural networks, especially architectures such as ResNet, which encode strong spatial inductive biases including locality and translation equivariance. In their influential paper *An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale*, Dosovitskiy et al. propose the Vision Transformer (ViT), showing that a pure transformer architecture can achieve state-of-the-art image classification performance without using any convolutional layers. Rather than processing an image through convolutional filters, ViT divides an image into fixed-size patches, linearly embeds each patch, adds positional embeddings, and treats the resulting sequence as analogous to a sequence of word tokens in natural language processing.

This paper is important because it demonstrates that the transformer architecture, originally developed for sequential data, can also serve as a general-purpose architecture for vision. At the same time, the paper makes a more subtle and important claim: the relative performance of ViTs and CNNs depends strongly on the amount of training data available**.** With limited data, convolutional networks tend to outperform transformers because their built-in inductive biases help them generalize more efficiently. With very large-scale pretraining, however, ViTs can match and eventually outperform strong convolutional baselines.

This project reviews the main technical content of the Vision Transformer paper, explains its connections to the material covered in the course, and presents a scaled-down empirical study comparing a small ViT and ResNet-20 on CIFAR-10 under varying training set sizes. In addition, we examine internal properties of the trained ViT through attention distance and positional embedding similarity visualizations, and conclude with possible extensions and applications of the method.

2. Main Technical Content and Contribution of the Paper

The core idea of the Vision Transformer is simple but powerful. An image of size $H \times W$is divided into non-overlapping patches of size $P \times P$. Each patch is flattened and projected into a learned embedding space. A learnable class token is prepended to the sequence, and positional embeddings are added to preserve spatial information. This sequence is then processed by a standard transformer encoder consisting of repeated blocks of multi-head self-attention and feedforward layers with residual connections and layer normalization. The final representation of the class token is passed to a linear classifier to produce the output class probabilities.

The technical contribution of the paper is not merely the adaptation of transformers to images, but the demonstration that a pure transformer architecture can be competitive with and eventually superior to CNNs for image recognition when trained at sufficient scale**.** Earlier vision models had used attention as an auxiliary component inside primarily convolutional

architectures. By contrast, ViT removes convolution entirely and applies the transformer directly to image patches.

A second major contribution is the paper's analysis of data regime dependence. The authors show that on relatively small datasets such as ImageNet, CNNs outperform ViTs of comparable scale, but as the pretraining dataset becomes much larger, such as ImageNet-21k and JFT-300M, ViTs close the gap and eventually surpass ResNet baselines. This result provides strong empirical support for the idea that architectural inductive bias and data scale can substitute for one another. CNNs perform well in low-data settings because they hard-code image structure, whereas transformers are more flexible but require more data to learn those same regularities.

The paper also provides interpretability evidence for this claim. It studies mean attention distance and shows that ViT attention becomes more global with layer depth, somewhat analogous to how receptive fields grow deeper in CNNs. It also visualizes learned positional embeddings and shows that, although they are learned as one-dimensional vectors, they recover meaningful two-dimensional spatial structure. These observations help explain how transformers can model image structure despite lacking convolutional inductive bias.

Overall, the paper is a landmark contribution because it establishes the transformer as a viable and eventually dominant architecture in computer vision, while also clarifying the role of data scale and inductive bias in modern deep learning.

3. Why the Paper Is Interesting in Relation to the Course

This paper connects strongly to several core themes of the course.

3.1 Connection to convolutional networks and computer vision

One of the major topics in the course is convolutional neural networks and their role in image classification. ViT is especially interesting because it directly challenges the assumption that convolutions are necessary for high-performance vision models. CNNs encode locality and translation equivariance, which are strong priors for images. ViT removes these priors and instead allows every patch to attend to every other patch from the first layer onward. This makes the ViT paper a natural extension of the CNN material covered in class.

3.2 Connection to attention and transformers

Another major course topic is attention mechanisms and transformers, especially in the context of language and sequence modeling. ViT is interesting because it shows that the same self-attention mechanism can be used beyond text. The architecture is fundamentally the same transformer encoder introduced for NLP, but applied to sequences of image patches instead of words. This demonstrates the generality of attention as a modeling primitive.

3.3 Connection to regularization, optimization, and data scale

The paper also relates to the course discussion of optimization and regularization. CNNs can be viewed as having strong implicit regularization because their architectural biases constrain the space of possible functions. Transformers have weaker built-in assumptions, which makes them more flexible but also more data-hungry. The ViT paper is therefore a clear example of how strong data scale can substitute for strong architecture-specific priors. This links naturally to broader course themes such as generalization, capacity, and the effect of training data size on model performance.

3.4 Broader conceptual relevance

Finally, the paper is interesting because it contributes to a larger trend in machine learning toward more general architectures trained at scale. This connects to the idea that progress increasingly comes from scalable, flexible methods combined with more data and computation, rather than from hand-crafted domain-specific architectures. ViT is one of the clearest examples of this trend within computer vision.

4. Experimental Implementation and Scaled-Down Replication

To test the paper's central claim in a setting feasible for a course project, we implemented a scaled-down experiment comparing a small Vision Transformer (ViT-S/4) and ResNet-20 on CIFAR-10.

4.1 Experimental setup

The dataset used was CIFAR-10. We trained both models from scratch using four fractions of the training set: 10%, 25%, 50%, and 100%. The Vision Transformer used a patch size of 4, which is suitable for the small 32×32 CIFAR-10 images. Both models were trained for 50 epochs using AdamW, cosine annealing, label smoothing, random crop augmentation, and random horizontal flipping.

The parameter counts were:

- **ViT-S/4:** 811,146 parameters
- **ResNet-20:** 272,474 parameters

Although the models were not parameter-matched, this comparison still allows us to examine how a small transformer and a small CNN behave under different data regimes.

4.2 Main classification results

The best test accuracies were:

Training fraction	ViT-S/4	ResNet-20
10%	0.5661	0.7260
25%	0.6779	0.8134

Training fraction	ViT-S/4	ResNet-20
50%	0.7536	0.8591
100%	0.8116	0.8873

These results show that ResNet-20 outperformed ViT-S/4 at every training fraction. However, the performance gap narrowed as more data became available. At 10% of the training data, the accuracy gap was approximately 16 percentage points, while at 100% of the training data the gap had narrowed to about 7.6 percentage points**.**

This is an important result because it reproduces the direction of the original ViT paper's claim, even if it does not reproduce the full crossover. In our small-scale CIFAR-10 setting, the CNN remains better throughout, but the transformer benefits more from increased data and steadily closes the gap.

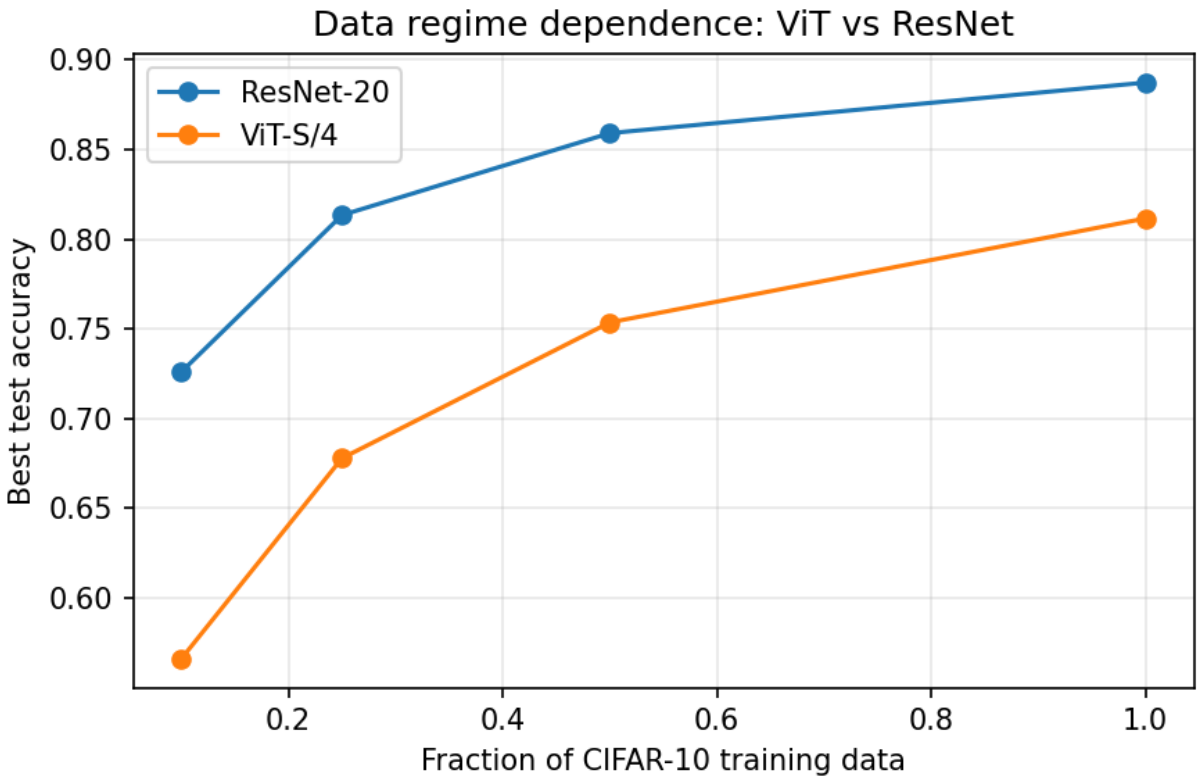

Figure 1. *Best test accuracy versus fraction of CIFAR-10 training data for ViT-S/4 and ResNet-20.*

4.3 Interpretation of scaling results

These results are consistent with the inductive bias explanation in the original paper. ResNet-20 has convolutional structure built in, which makes it more data-efficient. ViT-S/4 does not have

these same built-in assumptions and must instead learn image structure from data. This makes it less effective in low-data settings, but more responsive to increasing data volume.

It is important, however, not to overstate the result. Our experiment does not show that ViT overtakes ResNet. Instead, it shows a scaled-down version of the same pattern: CNNs dominate in the low-data regime, while transformers improve faster as more data becomes available. This is exactly the kind of simplified replication expected in a mini-project.

4.4 Attention distance analysis

We next examined one of the paper's interpretability ideas: mean attention distance by transformer layer. The computed values were:

- Layer 1: 15.36
- Layer 2: 12.46
- Layer 3: 15.29
- Layer 4: 15.28
- Layer 5: 15.63
- Layer 6: 15.61

The original paper reports that attention tends to become more global with depth. Our small model shows a broadly similar tendency, though not perfectly monotonic. Most layers exhibit relatively high mean attention distance, suggesting that even this small ViT attends over fairly broad spatial ranges. The second layer is noticeably more local, which may reflect a shallow-layer specialization or simply the noisier behavior expected in a smaller model trained from scratch on a limited dataset.

So, the correct interpretation is not that we replicated the exact monotonic curve from the large-scale paper, but that we found evidence of non-local attention and increasing globality in deeper layers, with some irregularity due to the small scale of the experiment.

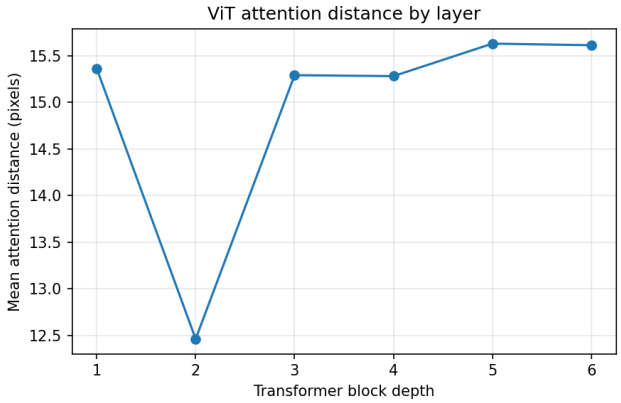

Figure 2. *Mean attention distance across transformer depth for the trained ViT-S/4 model.*

4.5 Positional embedding similarity analysis

We also visualized cosine similarity among learned positional embeddings. The original ViT paper reports that one-dimensional learned positional embeddings recover meaningful two-dimensional spatial relationships. Our results support that claim qualitatively.

The similarity maps show that query positions tend to have the most structured relationships with spatially nearby positions, and the center query in particular shows a relatively smooth, localized similarity pattern. Corner and edge queries display more asymmetric patterns, which is expected because they lie on image boundaries. Although the learned structure is not perfectly isotropic, the visualizations still suggest that the model has learned a meaningful notion of spatial layout.

This is significant because ViT does not build in 2D spatial structure through convolutions. Instead, that structure must emerge through training. Our results therefore provide a useful qualitative confirmation that even a small ViT can recover spatial organization through learned positional embeddings.

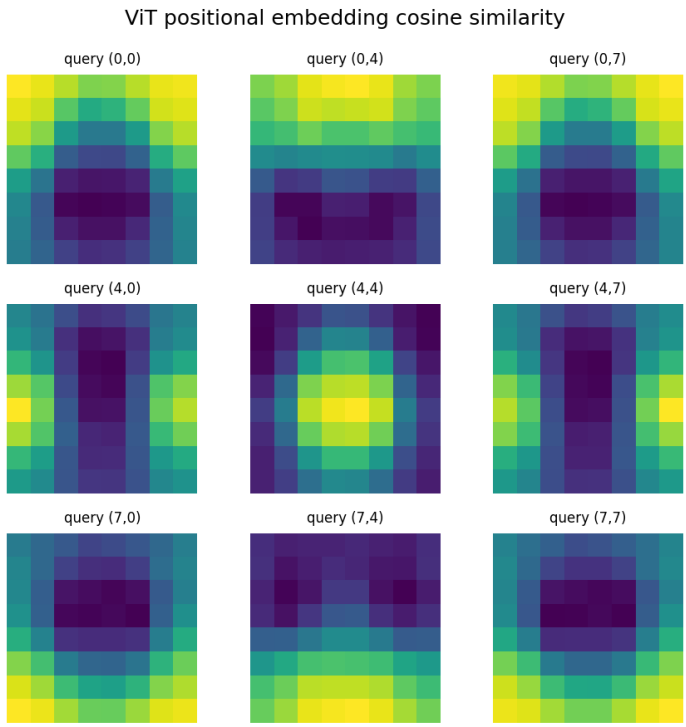

Figure 3. *Cosine similarity heatmaps for selected learned positional embeddings in ViT-S/4.*

4.6 Overall replication assessment

Taken together, the experiments partially replicate the key ideas of the original paper:

- ResNet performs better in the low-data regime.
- ViT improves more strongly as the amount of data increases.
- ViT develops non-local attention patterns.
- Learned positional embeddings recover spatial structure.

At the same time, the replication is necessarily limited. Our ViT does not surpass the ResNet baseline, the attention-distance pattern is only approximately consistent with the original paper, and the experiment is conducted on CIFAR-10 rather than on ImageNet-scale pretraining datasets. Still, for our course mini-project, this is a strong and meaningful scaled-down validation of the paper's main arguments.

5. Possible Methodological Extensions and Novel Applications

The Vision Transformer framework opens up several interesting directions for future work.

5.1 Improving data efficiency

One major extension is to improve ViT performance in smaller-data settings. Although the original ViT paper primarily shows that transformers become highly competitive at very large training scale, later work has focused on making them more data-efficient. In particular, DeiT demonstrates that Vision Transformers can perform much better in smaller-data regimes through stronger augmentation, knowledge distillation from a CNN teacher, and a more carefully tuned training recipe. A natural follow-up to our project would therefore be to implement a more data-efficient ViT training pipeline and test whether the gap with ResNet-20 narrows more quickly on CIFAR-10.

5.2 Reintroducing structured inductive bias

A second direction is to study architectures such as Swin Transformer, which add locality and hierarchical structure back into the transformer. This would allow a more direct comparison between pure transformers and hybrid vision architectures, and would help test whether partial inductive bias is the most effective compromise in moderate-data settings.

5.3 Self-supervised learning

Another important extension is self-supervised pretraining, especially masked autoencoders. Since transformers naturally process patches, they are especially well suited for patch-masking objectives. A project extension could pretrain a small ViT using masked reconstruction and then fine-tune it on CIFAR-10, testing whether self-supervised learning improves data efficiency.

5.4 Novel application beyond natural images

A particularly interesting novel application is to use ViT-style tokenization on non-image grid data. Any data that can be represented as a two-dimensional or spatiotemporal grid could potentially be processed using the same patch-and-attend principle. Examples include satellite data, urban demand heatmaps, emergency response time fields, traffic density maps, and other structured spatial surfaces. A future project could convert such data into patches and use transformer attention to model spatial dependencies between regions.

6. Conclusion

The Vision Transformer paper is a foundational contribution to modern machine learning because it demonstrates that image classification can be performed effectively using a pure transformer architecture. Its most important conceptual contribution is not only that transformers work for images, but that their relative advantage depends strongly on the amount of available training data. CNNs remain superior when data is limited because their inductive biases improve sample efficiency, while transformers become increasingly competitive as scale increases.

Our scaled-down CIFAR-10 experiments support this central insight. ResNet-20 outperformed ViT-S/4 at every training fraction, but the gap narrowed steadily as more data was used. Additional analyses of attention distance and positional embedding similarity further showed that the trained ViT learns broad spatial interactions and meaningful positional structure even without convolutional priors. Although our results do not reproduce the full large-scale crossover of the original paper, they successfully capture the main qualitative behavior.

More broadly, this project highlights a central lesson of modern deep learning: architectural bias, flexibility, and data scale are deeply interconnected. The Vision Transformer is compelling not only because it achieves strong empirical results, but because it clarifies how general-purpose architectures can replace domain-specific structure when sufficient data and computation are available.

References

Devlin, J., Chang, M. W., Lee, K., and Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL. arXiv:1810.04805.

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., and Houlsby, N. (2021). An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. ICLR. arXiv:2010.11929.

He, K., Zhang, X., Ren, S., and Sun, J. (2016). Deep Residual Learning for Image Recognition. CVPR. arXiv:1512.03385.

He, K., Chen, X., Xie, S., Li, Y., Dollár, P., and Girshick, R. (2022). Masked Autoencoders Are Scalable Vision Learners. CVPR. arXiv:2111.06377.

Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., and Guo, B. (2021). Swin Transformer: Hierarchical Vision Transformer using Shifted Windows. ICCV. arXiv:2103.14030.

Sutton, R. (2019). The Bitter Lesson. Incomplete Ideas blog.

Touvron, H., Cord, M., Douze, M., Massa, F., Sablayrolles, A., and Jégou, H. (2021). Training Data Efficient Image Transformers and Distillation Through Attention. ICML. arXiv:2012.12877.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., and Polosukhin, I. (2017). Attention Is All You Need. NeurIPS. arXiv:1706.03762.

Wightman, R. (2019). PyTorch Image Models. GitHub repository rwightman/pytorch-image-models.